Project: Worked examples
Title: Section properties
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Area, centroid, second moments of area (Ixx, Iyy, Ixy), principal axes and values, elastic and plastic section moduli, radii of gyration, torsion constant and (I-sections) warping constant for a parametric shape, an arbitrary polygon with holes, or a composite of placed parts. The x axis is horizontal (major axis of an I-section), y vertical.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Area and second moments of each outline, by Green's theorem | $A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$; $I_{xx} = \int y^2\,dA$; $I_{yy} = \int x^2\,dA$; $I_{xy} = \int xy\,dA$ |
| Second moments moved to the centroid by the parallel-axis theorem | $I_c = I - A\,d^2$ |
| Principal axes | $\tan 2\theta = -2I_{xy}/(I_{xx} - I_{yy})$ |
| Elastic section modulus | $W_{el} = I_{xx}/y_{max}$ |
| Plastic section modulus, about the equal-area axis found by bisection | $W_{pl} = \lvert Q_{above}\rvert + \lvert Q_{below}\rvert$ |
| Torsion constant by shape, a Saint-Venant series for a solid rectangle, $\pi d^4/32$ for a circle, Bredt for a closed thin-walled section, $\Sigma bt^3/3$ (plus a fillet term for rolled I) for an open section. | |
| Warping constant of an I-section | $I_w = I_{yy}\,(h - t_f)^2/4$ |

Basis: Properties of sections; torsion of solid and thin-walled sections[1](Appendix A, Table A.1; Ch. 10, Table 10.1); Torsional properties of rolled I-shapes[2](Appendix); Hot rolled steel channels, I and H sections: dimensions and masses[3]; Hot finished structural hollow sections: tolerances, dimensions and sectional properties[4].

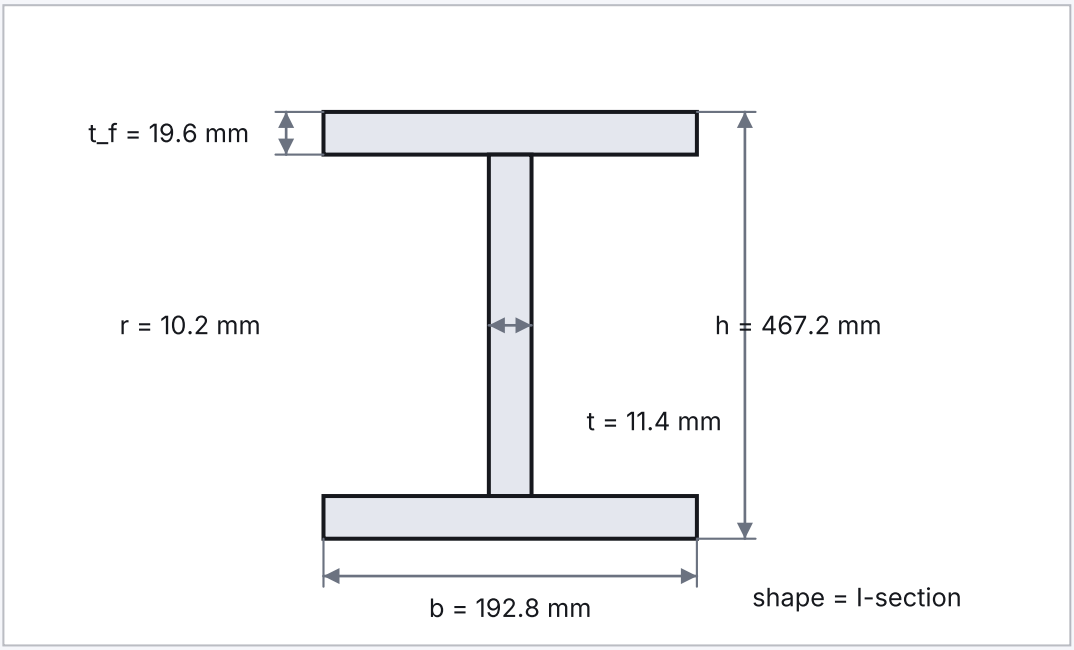



Figure 1 — A doubly symmetric I-section b by h, web t, flanges t_f, root radius r. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| shape | Shape | I-section | | spec: BS EN 10365 UB 457×191×98 | any |
| h | Depth h (or vertical leg) | 467.2 | mm | spec: BS EN 10365 UB 457×191×98 | 1 … 5000 mm |
| b | Width b (or horizontal leg) | 192.8 | mm | spec: BS EN 10365 UB 457×191×98 | 1 … 5000 mm |
| t | Wall / web / leg thickness t (t_w) | 11.4 | mm | spec: BS EN 10365 UB 457×191×98 | 0.1 … 500 mm |
| t_f | Flange thickness t_f | 19.6 | mm | spec: BS EN 10365 UB 457×191×98 | 0.1 … 500 mm |
| r | Root radius r (I, channel, tee, angle) or outer corner radius (hollow rectangle; blank = 1.5 t) | 10.2 | mm | spec: BS EN 10365 UB 457×191×98 | 0 … 500 mm |

## 3. Assumptions

A1. A polygon outline is simple (no self-intersection); holes lie inside the outline and do not overlap. (Green's theorem integrates the signed loops as given.)
A2. Rolled-section root fillets are quarter circles of radius r; flange tips are square; hollow-section corners are quarter circles of outer radius r_o (default 1.5 t) and inner radius r_i (default 1.0 t). (Published section tables use the same idealisation.)
A3. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

## 4. Working

[7] A = ½ ∮ (x dy − y dx) [1](Table A.1)
= Green's theorem over 108 vertices in 1 loop(s)
= 125.27 cm²
Area
[8] x_c = ∫x dA / A
= Green's theorem over 108 vertices in 1 loop(s)
= 0 mm
Centroid x (from the shape origin)
[9] y_c = ∫y dA / A
= Green's theorem over 108 vertices in 1 loop(s)
= 0 mm
Centroid y (from the shape origin)
[10] I_xx = ∫y² dA − A y_c²
= Green's theorem over 108 vertices in 1 loop(s)
= 45728 cm⁴
Second moment of area about the centroidal x axis
[11] I_yy = ∫x² dA − A x_c²
= Green's theorem over 108 vertices in 1 loop(s)
= 2347 cm⁴
Second moment of area about the centroidal y axis
[12] I_xy = ∫xy dA − A x_c y_c
= Green's theorem over 108 vertices in 1 loop(s)
= 0 cm⁴
Product of inertia about the centroid
[13] I_1 = (I_xx + I_yy)/2 + sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
= (45728 cm⁴ + 2347 cm⁴) / 2 + sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 45728 cm⁴
Major principal second moment
[14] I_2 = (I_xx + I_yy)/2 - sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
= (45728 cm⁴ + 2347 cm⁴) / 2 − sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 2347 cm⁴
Minor principal second moment
[15] theta = ½ atan2(-2 I_xy, I_xx - I_yy)
= ½ atan2(0, 43381)
= 0 deg
Angle from the x axis to the major principal axis (anticlockwise)
[16] y_top = y_max − y_c
= Green's theorem over 108 vertices in 1 loop(s)
= 233.6 mm
Extreme fibre above the centroid
[17] y_bot = y_c − y_min
= Green's theorem over 108 vertices in 1 loop(s)
= 233.6 mm
Extreme fibre below the centroid
[18] x_ext = max(x_max − x_c, x_c − x_min)
= Green's theorem over 108 vertices in 1 loop(s)
= 96.4 mm
Extreme fibre from the centroid, x
[19] W_el_x = I_xx / max(y_top, y_bot)
= 45728 cm⁴ / max(233.6 mm, 233.6 mm)
= 1957.5 cm³
Elastic section modulus about x (governing fibre)

[20] W_el_y = I_yy / x_ext
= 2347 cm⁴ / 96.4 mm
= 243.47 cm³
Elastic section modulus about y
[21] W_pl_x = |Q_above| + |Q_below| about the equal-area axis
= equal-area axis at y = 0 mm
= 2232.5 cm³
Plastic section modulus about x
[22] W_pl_y = |Q_right| + |Q_left| about the equal-area axis
= equal-area axis at x = 0 mm
= 378.9 cm³
Plastic section modulus about y
[23] i_x = sqrt(I_xx / A)
= sqrt(45728 cm⁴ / 125.27 cm²)
= 19.106 cm
Radius of gyration about x
[24] i_y = sqrt(I_yy / A)
= sqrt(2347 cm⁴ / 125.27 cm²)
= 4.3286 cm
Radius of gyration about y
[25] alpha_pl_x = W_pl_x / W_el_x
= 2232.5 cm³ / 1957.5 cm³
= 1.1404
Shape factor about x
[26] J = (2 b t_f³ + (h - 2 t_f) t_w³)/3 + 2 α D⁴ - 0.42 t_f⁴
= b = 192.8, t_f = 19.6, h = 467.2, t_w = 11.4, r = 10.2 mm
= 121.29 cm⁴
Torsion constant
[27] I_w = I_yy (h - t_f)² / 4 [2](DG9)
= 2347 cm⁴ × (467.2 - 19.6)² mm² / 4
= 1.1755 dm⁶
Warping constant

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| A | 125.27 cm² | Area |
| x_c | 0 mm | Centroid x |
| y_c | 0 mm | Centroid y |
| I_xx | 45728 cm⁴ | I_xx |
| I_yy | 2347 cm⁴ | I_yy |
| I_xy | 0 cm⁴ | I_xy |
| I_1 | 45728 cm⁴ | I_1 (major principal) |
| I_2 | 2347 cm⁴ | I_2 (minor principal) |
| theta | 0 deg | Principal axis angle |
| W_el_x | 1957.5 cm³ | W_el,x |
| W_el_y | 243.47 cm³ | W_el,y |
| W_pl_x | 2232.5 cm³ | W_pl,x |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| W_pl_y | 378.9 cm³ | W_pl,y |
| i_x | 19.106 cm | i_x |
| i_y | 4.3286 cm | i_y |
| alpha_pl_x | 1.1404 | Shape factor, x |
| J | 121.29 cm⁴ | Torsion constant J |
| I_w | 1.1755 dm⁶ | Warping constant I_w |

## 6. Checks (governing first)

No pass/fail checks in this calculation.

## 7. Applicability

WARNING — thin-walled-torsion: J: open thin-walled sum with the rolled-section fillet term.

## 8. Verification

Template struct.section-properties@1.0.1 Content hash b7d59144d05267d4 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Seaburg PA, Carter CJ. Torsional Analysis of Structural Steel Members. Steel Design Guide 9. American Institute of Steel Construction; 1997.
3. British Standards Institution. Hot Rolled Steel Channels, I and H Sections—Dimensions and Masses. BS EN 10365. BSI.
4. British Standards Institution. Hot Finished Steel Structural Hollow Sections—Part 2: Tolerances, Dimensions and Sectional Properties. BS EN 10210-2. BSI.